Yiquan Wang

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Education & Research Training

Xinjiang University 2023.09–Present
B.S. in Mathematics and Applied Mathematics, National Mathematics Base; expected graduation: 2027.06.
Mentors: Yan-Hong Qin and Kai Wei.

Tsinghua University & Shenzhen X-Institute 2024.06–Present
Joint Training, Zero One Scholar; expected completion: 2027.06.
Tsien Excellence in Engineering Program.

Shenzhen Bay Laboratory 2026.07–Present
Visiting Student, Kai Huang Research Group; Institute of Systems and Physical Biology.

Shenzhen Bay Laboratory 2025.07–2025.09
Visiting Student, Wen Yuan Research Group; Institute of Neurological and Psychiatric Disorders.

Professional Experience

Tsinghua University 2025.07–Present
Intern, Beijing Frontier Research Center for Biological Structure. Supervisors: Yafei Yuan and Tong Wang.
Protein and polypeptide design research; developed the Polypeptide Structure Database, Peptide Design Competition website, and Tong Wang Research Group website.

Institute of Software, Chinese Academy of Sciences 2024.09–2025.03
& Huawei MindSpore Community
Open Source Intern. Implemented VGG19-based Pollock-style generation with fractal and turbulent feature extraction.

Research Projects

Application of Artificial Intelligence in Whole Genome Selection Breeding 2025.12–2026.12

Urban Big Data, Multimodal Fusion and Spatial Intelligence Analysis 2025.11–2026.11

Conditional Diffusion Model on Copy Number Variation for Alzheimer's Disease Risk Assessment 2025.05–2026.12

HeDA: An Intelligent Agent System for Heatwave Risk Discovery through Automated Knowledge Graph Construction and Multi-layer Risk Propagation Analysis 2024.11–2025.09

From Signal to Symphony: Exploring 2D Sequence Representations for Protein Function Prediction 2024.08–2025.10

Research on the Generation Cyclability of Cartesian Product Graphs and Related Problems 2024.03–2025.06

Conferences & Additional Training

35th Chinese Chemical Society Congress, Chongqing, China 2026.04

Summer Camp, Tsinghua University–Peking University Center for Life Sciences 2025.07

AI Winter School, Brown University Department of Physics 2025.01

Artificial Intelligence and Technology Ethics Training Course, Chinese Association for Artificial Intelligence 2024.09–2024.12

Summer School of Mathematical Logic, Fudan University 2024.08

Discussion Class, National Tianyuan Mathematics Central Center, Wuhan University 2024.03–2024.06

Selected Publications

† *Co-first author;* * *corresponding author.* Full publication list

1. Representation Learning & Biomolecular Mechanisms

[1] **Wang, Y.**, Cai, M., Ma, Y., Shi, T.*, & Wei, K.* (2026). **Sequence context decodes multi-state conformational heterogeneity from crystallographic B-factors**. *Science China Life Sciences*.

[2] **Wang, Y.*** & Wei, K.* (2026). Contrasting binding architectures underlying the asymmetric M327I resistance of menin inhibitors. *Journal of Molecular Modeling*.

[3] **Wang, Y.*** (2026). Comment on "Protein language visualizer: a repository for homology exploration with language model embeddings". *Digital Discovery*.

[4] **Wang, Y.**†, Cai, M.†, Ma, Y., Tellier, A.*, & Wei, K.* (2026). Integrating deep learning and pangenomics to recover missing heritability from wild structural variations. *BMC Genomics*.

[5] **Wang, Y.*** & Cai, M. (2026). Integrative transcriptomics, machine learning, and molecular docking derive a DAM-like macrophage signature for risk stratification and therapeutic nomination in glioblastoma. *Computational Biology and Chemistry*.

[6] **Wang, Y.*** & Huang, T. Y.* (2026). Comment on: "GLM7 – A Novel Composite Glycolipid Index Derived from Routine Health Indicators for Enhanced Diagnosis and Prediction of Multimorbidity". *Advanced Science*.

[7] **Wang, Y.**, Cai, M., Dong, Y., Ma, Y., & Wei, K.* (2025). **From Signal to Symphony: Exploring 2D Sequence Representations for Protein Function Prediction**. *Journal of Chemical Information and Modeling*.

[8] **Wang, Y.**†, Ma, Y.†, Chang, Y., Yan, J., Zhang, J., Cai, M., & Wei, K.* (2025). Diffusion Models at the Drug Discovery Frontier: A Review on Generating Small Molecules versus Therapeutic Peptides. *Biology*.

[9] Wang, X.†*, **Wang, Y.**†*, Huang, T. Y., Dong, Y., Deng, J., Xu, L., Li, X., & He, R. (2025). Octopus Inspired Optimization (OIO): A Hierarchical Framework for Navigating Protein Fitness Landscapes. *2025 IEEE International Conference on Bioinformatics and Biomedicine*.

2. First-Principles of Biological Complexity Based on Mathematical Physics

[10] **Wang, Y.*** (2026). Spectral analysis of protein backbone geometry reveals abrupt helix–coil boundaries. *The European Physical Journal E*.

[11] **Wang, Y.*** (2026). **Descriptive power and predictive limits of a discrete Hasimoto–DNLS model of protein backbone structure**. *Physica D: Nonlinear Phenomena*.

3. Interdisciplinary Explorations & Collaborative Synergies

[12] **Wang, Y.**, Zai, J., Liu, Z., Chen, J., & Sabir, E.* (2026). Hamiltonian Connectivity in k-Ary n-Cubes under a Region-Based Fault Model. *Information Sciences*.

[13] **Wang, Y.***, Huang, T. Y., Gao, Q., Chang, Y., & Zhang, J. (2026). AI Driven Discovery of Bio Ecological Mediation in Cascading Heatwave Risks. *Physics and Chemistry of the Earth, Parts A/B/C*.

[14] **Wang, Y.***, Chang, Y., Zhang, J., Cai, M., Yan, J., & Zhao, H.*, for the Alzheimer's Disease Neuroimaging Initiative (2026, accepted). Geometric Fragility in Alzheimer's Disease: Probing the Loss of Hippocampal Hierarchical Abstraction via Contrastive Point Cloud Modeling. *2026 IEEE International Conference on Systems, Man, and Cybernetics*.

Awards & Honors

China Undergraduate Life Science Contest: National Second Prize 2026

China International College Students' Innovation Competition: Provincial Gold Prize 2026

"Challenge Cup" China College Students' Extracurricular Academic Science and Technology Works Competition: Autonomous Region Second Prize 2026

China College Students' E-Commerce "Innovation, Creativity and Entrepreneurship" Challenge: Autonomous Region Second Prize 2026

UN Global Leadership and ESG Programme / X-Institute Shenzhen "AI for the Young and the Elderly" Challenge: Best Innovation Award 2026

China College Student Self-improvement Star 2025

Contemporary Undergraduate Mathematical Contest in Modeling: National Second Prize 2025

Smart Chemical City Challenge (Xinjiang Station): Second Prize 2025

SynBio Challenges: Two Silver Awards 2025

X-Fusion "Global Innovator Fusion Conference": Best Poster Award 2025

Mathematical Contest in Modeling: Honorable Mention 2025

Alibaba Cloud Tianchi University Student Competition: National Finals, 17th Place 2024

Asia-Pacific Mathematical Modeling Competition: National Third Prize 2024

Academic Service

Conference reviewer: 2026 IEEE International Conference on Systems, Man, and Cybernetics.

Journal reviewer

Current Science
Engineering Research Express
European Journal of Physics
F1000Research
Frontiers in Aging Neuroscience
Frontiers in Bioinformatics
Frontiers in Chemistry
Frontiers in Complex Systems
Frontiers in Immunology
Frontiers in Physics
Mini-Reviews in Medicinal Chemistry
Physics and Chemistry of the Earth, Parts A/B/C
Physics Open
Urban Climate

Workshop reviewer: Workshops at the International Conference on Learning Representations, International Conference on Machine Learning, and Conference on Neural Information Processing Systems (2025–2026).

Conference staff: SMART Genome Symposium: From Structure to Function, Shenzhen Medical Academy of Research and Translation, Shenzhen, China, 2026.10.15–18.

Memberships: IEEE Biometrics Council; Chinese Chemical Society; Biophysical Society of China.